02443 Stochastic Simulation

Day 3: Discrete event simulation (The event-by-event principle)

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Agenda

1. Introduction
2. Queueing systems
3. Subsampling

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Discrete event simulation

Characteristics

- Continuous but asynchronous time.
- Systems described by discrete state variables.

Examples

- Inventory systems
- Communication systems
- Traffic systems (simple models)

The simulation is driven by the *event-by-event principle*.

Elements of a discrete event simulation

A discrete event simulation program typically consists of the following components.

Core elements

- Real-time clock
- State variables
- Event list(s)
- Statistical accumulators

Together, these components describe the current state of the system and control the progression of the simulation.

The event-by-event principle

The simulation proceeds by repeatedly processing the next scheduled event.

Algorithm

1. Advance the simulation clock to the next event.
2. Invoke the corresponding event-handling routine.
3. Collect relevant statistics.
4. Update the system state variables.
5. Generate and schedule future events by inserting them into the event list(s).
6. Return to Step 1.

Analysing steady-state behaviour

When studying long-run performance, special care must be taken during the initial phase of the simulation.

Burn-in period

- The simulation often starts in an artificial initial state.
- Observations from the initial period may therefore be biased.
- The length of the burn-in period typically has to be determined experimentally.

Precision of estimates

- Confidence intervals can be constructed using sub-samples.
- Variances can be estimated from independent or approximately independent replications.

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Introduction to queueing systems

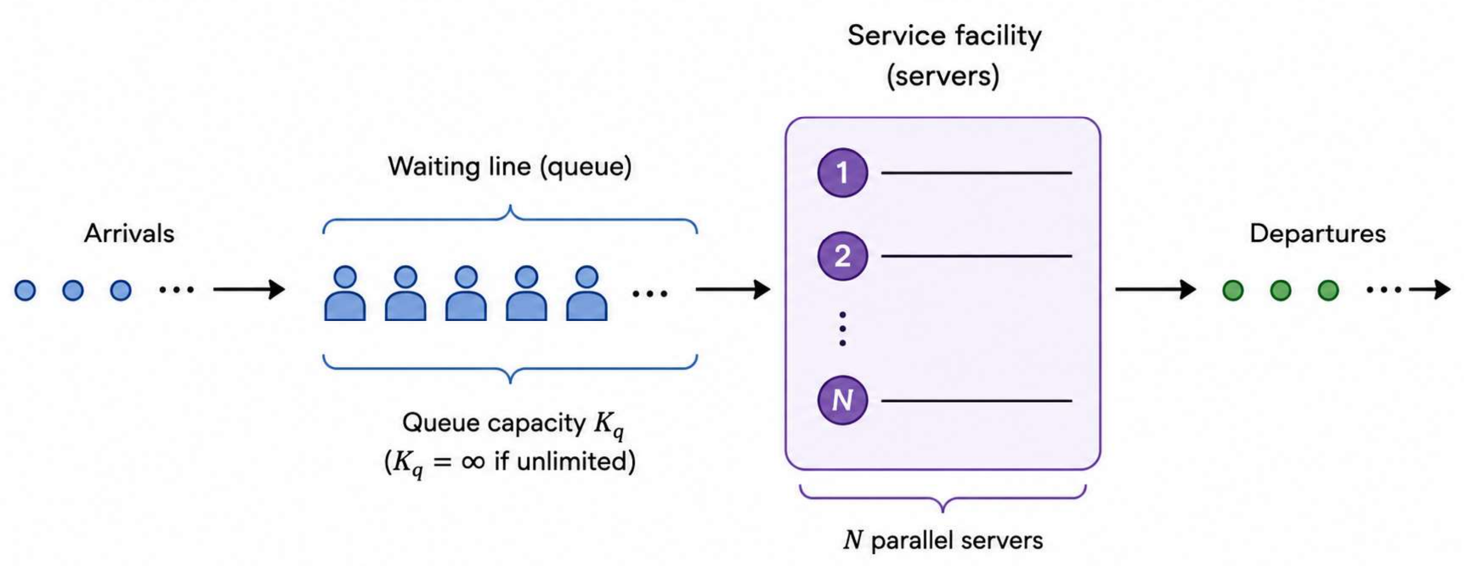
Many discrete-event simulations involve queueing systems.

Important components

- Arrival process
- Service-time distribution(s)
- Service unit(s)
- Priorities
- Queueing discipline

Different choices of these components lead to different queueing models and system behaviour.

Illustration of a simple queueing systems



Credit: ChatGPT

Mathematical model of a simple queueing systems

A queueing system is typically characterized by the following components.

Arrival process $A(t)$ - describes the arrival pattern of customers to the system.

Service process $S(t)$ - describes the service-time distribution.

Relevant system characteristics

- Finite or infinite waiting room
- One or more servers

Kendall notation
Queueing systems are commonly described using $A(t)/S(t)/N/K$, where

- N denotes the number of servers,
- K denotes the capacity of the system.

In some references, K refers only to the waiting room capacity.

Performance measures

The purpose of a queueing simulation is typically to estimate one or more performance measures.

Waiting-time distribution

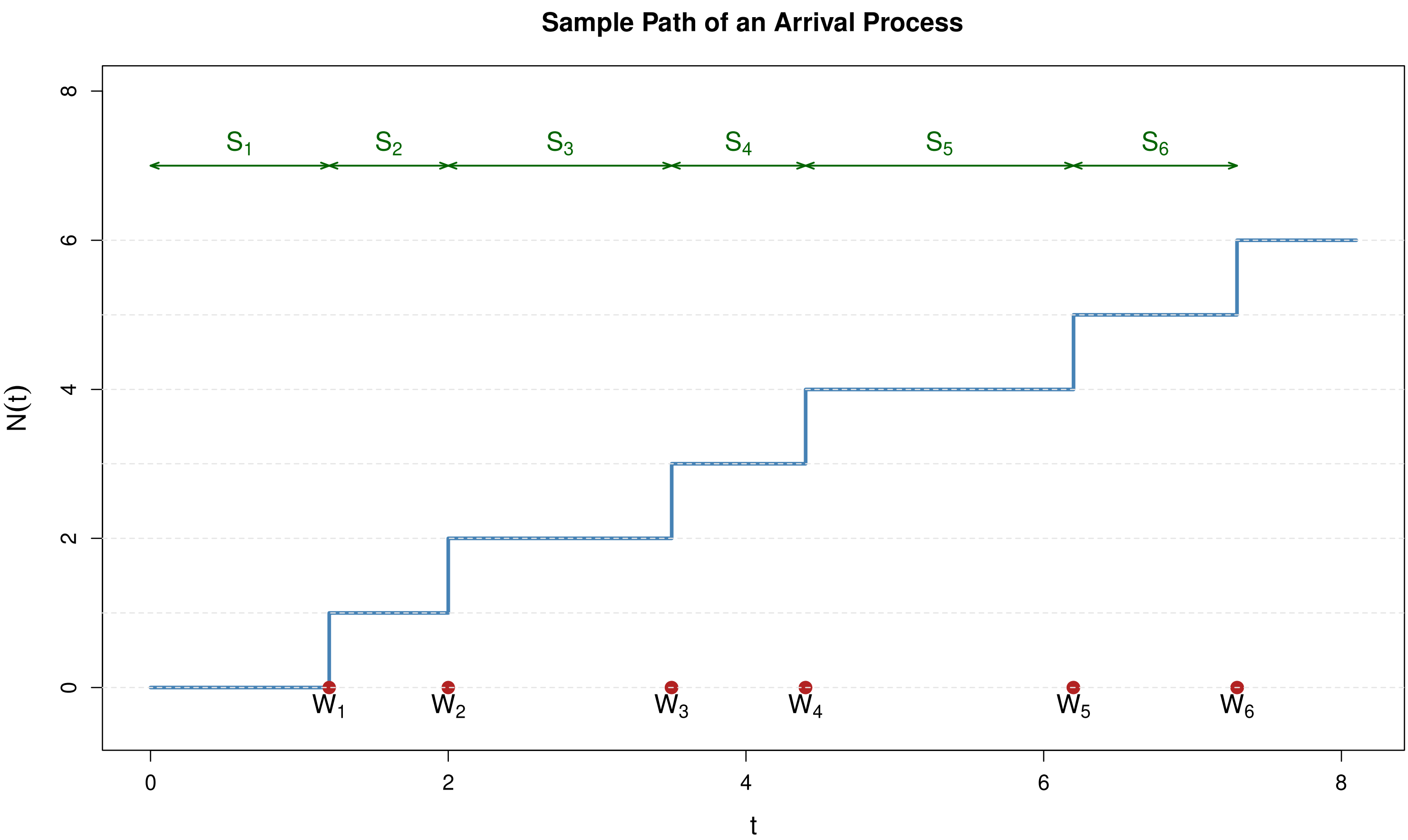
- Mean waiting time
- Variance of waiting time
- Quantiles

Other performance measures

- Blocking probabilities
- Utilization of equipment (servers)
- Queue-length distribution

These measures help quantify the performance and efficiency of the system under consideration.

Example of an arrival process



The Poisson process

The Poisson process is the most common choice of arrival process in queueing models.

Characteristics

- The Poisson process has independent and exponentially distributed inter-arrival times (sojourn times): $S_i \sim \exp(\lambda)$.
- The number of arrivals in a Poisson process in non-overlapping (disjoint) time intervals are independent.
- The number of arrivals in a Poisson process up to time t is Poisson distributed: $N_t = N(t) = \text{Poisson}(\lambda t)$

If the inter-arrival times are independent but follow a general distribution, the process is called a *renewal process*.

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Motivation

Consider an evolving system, e.g., a discrete-event simulation or a queueing system. Suppose we are interested in estimating some performance measure of the stochastic system.

The simulation produces an estimate and, during the simulation, we can track cumulative quantities. However, based on a single simulation run, it is generally difficult to assess the uncertainty of the estimator.

Possible solutions:

1. Repeat the simulation multiple times.
2. **Apply subsampling.**

Basic idea: Divide the simulation output into sub-samples (batches) and compute an estimate from each sub-sample. The variability between these estimates can then be used to estimate the variance of the overall estimator.

Sub-samples and precision of estimates

Important remark:
There exist numerous subsampling strategies designed to improve validity for different types of systems. The simplest approach is to divide the simulation output into batches of consecutive observations.

Important considerations

- The sub-samples should be independent whenever possible.
- The dependence structure in the sub-samples should be considered.

Precision
For independent sub-samples, the standard deviation of the estimator decreases proportionally to

$$\frac{1}{\sqrt{n}},$$

where n denotes the number of sub-samples.

Confidence intervals based on sub-samples

Suppose we wish to estimate a parameter θ. Assume that we obtain n approximately independent estimates:

$$\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_n.$$

Sample mean

$$\bar{\theta} = \frac{1}{n}\sum_{i=1}^{n} \hat{\theta}_i.$$

Sample variance

$$S_\theta^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} \hat{\theta}_i^2 - n\bar{\theta}^2\right).$$

Confidence intervals based on sub-samples (continued)

Motivated by the Central Limit Theorem, an approximate $(1-\alpha)$-confidence interval for θ is

$$\left[\bar{\theta} + \frac{S_\theta}{\sqrt{n}} t_{\alpha/2}(n-1), \bar{\theta} + \frac{S_\theta}{\sqrt{n}} t_{1-\alpha/2}(n-1)\right].$$

Here, $t_p(n-1)$ denotes the p-quantile of the t-distribution with $n-1$ degrees of freedom.

When the number of sub-samples is large, the t-distribution may be approximated by the standard normal distribution.

$$\left[\bar{\theta} + \frac{S_\theta}{\sqrt{n}} u_{\alpha/2}, \bar{\theta} + \frac{S_\theta}{\sqrt{n}} u_{1-\alpha/2}\right],$$

where u_p denotes the p-quantile of the standard normal distribution.